

NAKUL S

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ACADEMIC QUALIFICATION

- **Bangalore Institute of Technology, Visvesvaraya Technological University** **Jul'19 – Feb'23**
Bachelor of Engineering in Computer Science, *CGPA: 8.1*
Achievement:
- Secured a position among the top 10% of candidates in the **National Standard Examination in Chemistry** (NSEC) 2018-19, conducted by the Indian Association of Physics Teachers, out of a total of 42,443 participants nationwide.
 - Stood 2nd in the 2022 Debate Competition on *"Is Artificial Intelligence a Threat to Human Employment?"* among 20+ teams

ACADEMIC PROJECTS

- **City Distance Optimization Using Dijkstra's Algorithm (DSA Mini Project) | Team size: 2** **Mar'22-May'22**
- Applied *Dijkstra's* algorithm in *C* to determine the shortest path between cities in a graph-based model, showcasing its potential for routing and logistics
 - Designed an efficient and scalable solution for *city-to-city* distance calculation, relevant for transportation systems
- **Art Gallery Management Platform | Team size: 2** **Mar'22-May'22**
- Developed a web-based platform to streamline art gallery management, enabling artists and curators to easily upload, organize, and showcase artwork through an intuitive interface
 - Structured a comprehensive *MySQL* database to manage artwork inventory, track sales, and ensure data accuracy, supporting smooth gallery operations
 - Implemented *PHP* for server-side functionality, along with *HTML5* for content organization and *CSS3* for styling, resulting in a responsive and visually engaging user experience
 - Automated artwork display and inventory updates, reducing manual workload by **40%** and significantly increasing operational efficiency for gallery staff
- **Machine Learning-Based Credit Card Fraud Detection Team size: 3** **Aug'22-Oct'22**
Tools Used: Python, Scikit-learn, Pandas, NumPy, Matplotlib, Jupyter Notebook
- Constructed a machine learning model to identify fraudulent credit card transactions by analyzing transaction patterns and behaviors, amplifying financial security
 - Carried out *data preprocessing* and *feature engineering* to optimize input data quality, bolstering the model's performance and reliability
 - Built and tested multiple classification models, including *Decision Trees*, *Random Forest*, and *Logistic Regression*, to determine the best-performing algorithm for fraud detection
 - Tuned model for precision and recall to effectively reduce false positives and false negatives, achieving a **95%** accuracy rate and minimizing false-positive alerts by **20%**
- **Speech Emotion Recognition Using Machine Learning | Team size: 4** **Oct'22-Nov'22**
Tools Used: Python, TensorFlow, Keras, Librosa, Pandas, NumPy, Scikit-learn
- Built a machine learning model to classify emotions from speech, focusing on detecting emotions like happiness, anger, sadness, and neutrality
 - Extracted essential audio features, including *Mel-frequency cepstral coefficients (MFCCs)*, to accurately capture and represent vocal characteristics
 - Employed *Support Vector Machines (SVM)* and *Neural Networks* for thorough emotion classification, optimizing model performance for audio-based emotion recognition
 - Achieved **87%** accuracy in classifying emotions from speech inputs, making the model suitable for real-world utilization in call centers, mental health monitoring, and smart virtual assistants
- **Stock Price Prediction Using CNN and LSTM | Team size: 4** **Oct'22-Feb'23**
Tools Used: Python, Jupyter Notebook, Google Colab
- Developed a predictive model for forecasting stock prices utilizing *Convolutional Neural Networks (CNN)* and *Long Short-Term Memory (LSTM)* networks, effectively combining spatial and temporal data analysis
 - Analyzed historical stock data to identify spatial patterns through CNN and capture temporal dependencies with LSTM, significantly improving prediction accuracy for future stock prices
 - Conducted extensive data preprocessing, including handling missing values and normalizing data, to ensure high-quality input for the model
 - Attained a **15%** improvement in prediction accuracy compared to traditional models such as *ARIMA* and simple moving averages
 - Leveraged *Python* and *Jupyter Notebook* for model development and data analysis, while using *Google Colab* for efficient collaboration and resource management

PERSONAL PROJECT

➤ **Responsive Website Development for Modular Furniture Company**

Mar'20-Jun'20

Tools Used: HTML5, CSS3, JavaScript

- Designed a fully responsive and user-friendly website for Styllent Interiors, a modular furniture company to boost their online visibility and customer engagement
- Utilized **HTML5**, **CSS3**, and **JavaScript** to create an interactive platform that effectively showcases a diverse range of furniture products and designs
- Increased the company's web traffic by **30%** that led to higher product inquiries and optimized loading times

PROFESSIONAL EXPERIENCE

➤ **Software Developer I, Ellucian Higher Education Systems**

Bangalore, India | Jun'23 – Present

Tools Used: React, Typescript, NestJs, MySQL

- Developed and implemented **Playwright test automation scripts**, successfully identifying and fixing **UI bugs** to enhance application performance and user experience
- Conducted **unit testing** on **eTranscript APIs** for the **Send Modernization** project, achieving coverage across complex features and ensuring high code quality
- Analyzed **Datadog logs** to troubleshoot and resolve **503 errors** to make the system reliable
- Participated in **California Critical Scenarios Testing** for the March release on the **Oracle environment** and executed **California Regression Testing** on **PGDEV01**, validating system stability and performance after updates

Key Project: Grails 5 to Grails 6 Migration for eTranscript API

Team size: 2

- Led the migration of the **eTranscript API** from **Grails 5** to **Grails 6**, ensuring compatibility with the latest framework features and significantly upgrading performance and security
- Updated dependencies and refactored **Groovy code**, implementing best practices for code maintainability and readability
- Conducted thorough testing to guarantee seamless API functionality post-migration, achieving a successful transition with **zero downtime**
- Improved system performance by **15%** and enhanced maintainability, enabling the adoption of new **Grails** features that streamlined **API development** and testing workflows
- Utilized tools including Groovy, Grails, IntelliJ IDEA, Jenkins, JUnit, and Git to facilitate efficient development and deployment processes

Achievement: Recognized as the **'Top-Performing Member'** among a 5-member team in the second quarter

➤ **Technical Intern, Ellucian Higher Education Systems**

Bangalore, India | Feb'23 – May'23

- Contributed to the development of a key feature in the **Ellucian Document Management system**, which significantly streamlined the document storage and retrieval process for better user efficiency
- Designed and implemented **APIs** for smooth coordination between the frontend and backend, improving data handling and user experience
- Developed dashboard pages using **React** and **TypeScript**, resulting in a more intuitive interface that reduced document retrieval time by **15%**
- Created unit tests to ensure **100%** code coverage for the new interface, maintaining high code quality and contributing to a smooth, error-free user experience

Key Project: Student Academic Performance Tracking System (Hackathon Project)

- Developed a user-friendly system for tracking student academic performance using **React JS** for an intuitive front-end interface
- Utilized **Data Connect** for secure data retrieval and management and employed the **Ellucian Path Design System** to maintain a consistent user interface across the platform
- Achieved a **25%** reduction in errors related to **Grade Point Average (GPA)** and **credit tracking**, providing real-time updates to the faculty

Achievement: Secured **runner-up** position in the Hackathon event for delivering a practical and scalable solution

➤ **Web Developer Intern, Suvidha Foundation**

Nagpur, Maharashtra, India | Sep'21 – Nov'21

- Redesigned the login page and implemented this interface using **HTML**, **CSS**, and **JavaScript** for an internal web application using Figma to improve the page's usability and visual appeal
- Reduced the login times by **20%** which resulted in higher user satisfaction and streamlined webpage performance

ADDITIONAL INFORMATION

- **Leadership:** *Captain of College Cricket Team* – Spearheaded the team to victory in numerous inter-school and inter-college cricket tournaments through proper team management and arranging training schedules
- **Workshops:** *Android App Development*, Bangalore Institute of Technology (Sep'22) | *Web Development Bootcamp*, Bangalore Institute of Technology (Mar'22)
- **Certifications (Coursera):** *Interactivity with JavaScript*, UMich | *Python Data Structures*, UMich | *Introduction to HTML5*, UMich | *Introduction to CSS3*, UMich | *Using Databases with Python*, UMich | *Using Python to Access Web Data*, UMich | *C for Everyone: Programming Fundamentals*, UC Santa Cruz | *AWS S3 Basics* | *Oracle Database Foundations*

- **Community Service:** Taught English and Math to **70-90** class VII to IX government school students | Assisted in meal preparation and distribution for underprivileged families residing in slums | Donated blood during community blood donation camps
- **Other Technical Skills:** Java, C++, Windows, MacOS, MS Office (Word, Excel, Access, PowerPoint, TypeScript, Node